Application Note

ggwas: a comprehensive ggplot2 toolkit for genome-wide association study visualization

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Abstract

Summary: Genome-wide association studies (GWAS) are central to modern genetics, yet the R ecosystem lacks a unified, customizable visualization package that scales to biobank-era datasets. We present ggwas, a ggplot2-based R package providing 17 publication-ready plot types for GWAS summary statistics. Beyond standard Manhattan and QQ plots, ggwas offers novel visualizations including enrichment Manhattan plots with functional annotation overlays, multi-trait Manhattan overlays for pleiotropy detection, PheWAS plots, colocalization displays, fine-mapping credible set figures, genetic correlation heatmaps, genome-wide signal heatmaps, SNP density karyograms, and a dual-track density–signal comparison for quality control. Smart downsampling enables interactive-speed rendering of datasets with millions of variants (up to 9$\times$ faster than existing tools). Built-in journal themes, colorblind-safe palettes, and gene annotation streamline the path from analysis to publication.

Availability and Implementation: ggwas is implemented in R and available under the MIT license at `https://github.com/bczech/ggwas`. Documentation: `https://bczech.github.io/ggwas/`. DOI: 10.5281/zenodo.20815110.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Genome-wide association studies have become the primary approach for mapping genetic variants underlying complex traits and diseases, with recent biobank-scale studies testing millions of variants across hundreds of thousands of individuals (Bycroft *et al.*, 2018; Kurki *et al.*, 2023). The resulting summary statistics require effective visualization at every stage of analysis: quality control, discovery, replication, and biological interpretation.

The most widely used R package for GWAS visualization, `qqman` (Turner, 2018), provides Manhattan and QQ plots using base R graphics. While widely adopted (cited >2 000 times), it offers limited customization and does not integrate with the ggplot2 ecosystem (Wickham, 2016) that has become the standard for scientific graphics in R. `CMplot` (Yin, 2024) extends the repertoire to circular Manhattan plots and SNP density displays but also relies on base R graphics with a restrictive API. The interactive `manhattanly` package was removed from CRAN in June 2025 due to unresolved maintenance issues. The `locuszoomr` package (Lewis *et al.*, 2025) provides ggplot2-based regional association plots but is limited to locus-level visualization without genome-wide plot types.

Beyond traditional GWAS plots, modern post-GWAS workflows demand specialized visualizations that researchers currently construct through ad hoc scripts: PheWAS plots for biobank phenome scans (Denny *et al.*, 2013), colocalization displays for integrating GWAS with eQTL data (Giambartolomei *et al.*, 2014), fine-mapping credible set figures showing posterior inclusion probabilities from SuSiE (Wang *et al.*, 2020), and genetic correlation matrices from LD Score regression (Bulik-Sullivan *et al.*, 2015). The absence of standardized tools leads to inconsistent styling across publications and substantial duplicated effort. We developed ggwas to address these gaps with a single, comprehensive, ggplot2-native package.

2 Features

2.1 Plot types

The package provides 17 visualization functions organized into four categories (Table 2; Fig. 1; Supplementary Fig. S1):

Core GWAS plots. The Manhattan plot supports smart downsampling, customizable significance thresholds, a broken y-axis mode (`y_truncate`) for studies with extreme p-values, SNP highlighting, and automatic labeling of top hits. The QQ plot includes 95% confidence bands from the beta distribution, reports the genomic inflation factor (λ_{GC}), and supports stratification by minor allele frequency. The Miami plot mirrors two Manhattan plots for discovery–replication comparison. The locus zoom plot shows regional association with optional linkage disequilibrium coloring and gene annotation tracks.

Novel genome-wide visualizations. The genome-wide p-value heatmap bins variants by position and chromosome, providing a compact overview of association signals without downsampling through pre-aggregation. The effect-size volcano displays $\hat{\beta}$ against $-\log_{10}(p)$. The circular Manhattan uses a circos-style layout with multi-ring support for trait comparison. The enrichment Manhattan overlays functional genomic annotations (enhancers, promoters, eQTLs) from ENCODE (ENCODE Project Consortium, 2012) or Roadmap Epigenomics (Roadmap Epigenomics Consortium, 2015). The multi-trait Manhattan overlays results from multiple GWAS on a shared genomic axis, with automatic detection of pleiotropic variants.

Post-GWAS visualizations. The PheWAS plot arranges phenotype associations by category with directional effect indicators. The colocalization plot displays two association signals (e.g., GWAS and eQTL) on a shared coordinate axis. The fine-mapping plot encodes posterior inclusion probability (PIP) from SuSiE (Wang *et al.*, 2020) or FINEMAP (Benner *et al.*, 2016) as point size, with credible set membership indicated by color. The genetic correlation matrix displays trait–trait correlations from LDSC (Bulik-Sullivan *et al.*, 2015) as a clustered heatmap. The genetic architecture plot visualizes minor allele frequency versus effect size, revealing the polygenic or oligogenic nature of a trait.

Genome-wide density and QC. The SNP density karyogram visualizes variant distribution across chromosomes in two styles: binned heatmap and scatter-on-chromosome-outline. Centromere positions are marked when species-specific chromosome data is provided (built-in for human hg38, mouse mm39, and cattle ARS-UCD1.2; any UCSC assembly via `chr_info_ucsc()`). The density–signal comparison pairs a density track with an association signal track for each chromosome, enabling detection of regions where strong signals coincide with unusually high variant density—a hallmark of genotyping artifacts.

2.2 Data handling and annotation

The package natively reads output from PLINK (Purcell *et al.*, 2007), REGENIE (Mbatchou *et al.*, 2021), GCTA (Yang *et al.*, 2011b), and GEMMA (Zhou and Stephens, 2012), with automatic column name detection. A generic reader (`read_gwas_table()`) enables one-line loading of results from any tool. Gene annotation (`manhattan_genes()`) maps lead SNPs to nearest genes with ggrepel-based overlap avoidance for publication-ready labeling.

2.3 Customization

All functions return standard ggplot objects. Six journal themes match respective figure style guides (Nature, Science, Cell, PLOS for publications; Presentation and Poster for talks), with runtime font detection for cross-platform compatibility. Fourteen colorblind-safe palettes and publication presets are included.

2.4 Performance

Smart downsampling preserves all significant variants ($p < 0.001$) while spatially binning the non-significant background. Benchmarking on the GIANT height GWAS (Yengo *et al.*, 2022) (1.37M variants; Table 1) shows 2–9× speedup over `qqman`, with the advantage increasing as dataset size grows.

Table 1. Manhattan plot rendering time (seconds) on GIANT height GWAS data (Yengo *et al.*, 2022).

Variants	qqman	ggwas	Speedup
50k	0.17	0.06	2.6×
200k	0.80	0.11	7.0×
500k	2.03	1.01	2.0×
1M	4.08	1.24	3.3×
1.37M (full)	5.71	0.65	8.8×

Table 2. Feature comparison with existing packages.

Feature	qqman	CMplot	ggwas
ggplot2-native	–	–	✓
Manhattan + QQ	✓	✓	✓
Miami / Locus zoom	–	–	✓
Circular Manhattan	–	✓	✓
Enrichment overlay	–	–	✓
Multi-trait	–	–	✓
PheWAS	–	–	✓
Colocalization	–	–	✓
Fine-mapping (PIP)	–	–	✓
Genetic correlation	–	–	✓
SNP density	–	✓	✓
Density vs signal	–	–	✓
Journal themes	–	–	✓
Smart downsampling	–	–	✓

3 Conclusion

The ggwas package provides a comprehensive, ggplot2-native visualization toolkit for GWAS. By covering the full workflow from quality control through discovery to post-GWAS interpretation, it eliminates the need for ad hoc plotting scripts. Novel visualizations such as the enrichment Manhattan, genome-wide heatmap, multi-trait overlay, and density–signal dual-track comparison are not available in existing packages. The SNP density karyogram with built-in chromosome data for human, mouse, and cattle—and UCSC integration for any other species—makes the package applicable across organisms. Performance optimizations make ggwas practical for biobank-scale datasets.

Supplementary Data

Supplementary Figure S1 demonstrates all 17 plot types on GIANT height GWAS data: (S1) Manhattan, (S2) QQ, (S3) Miami, (S4) locus zoom, (S5) circular Manhattan, (S6) enrichment Manhattan, (S7) multi-trait Manhattan, (S8) genome-wide heatmap, (S9) effect-size volcano, (S10) summary dashboard, (S11) PheWAS, (S12) colocalization, (S13) fine-mapping, (S14) genetic correlation, (S15) genetic architecture, (S16) SNP density karyogram, (S17) density–signal comparison.

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Conflict of Interest

The author declares no competing interests.

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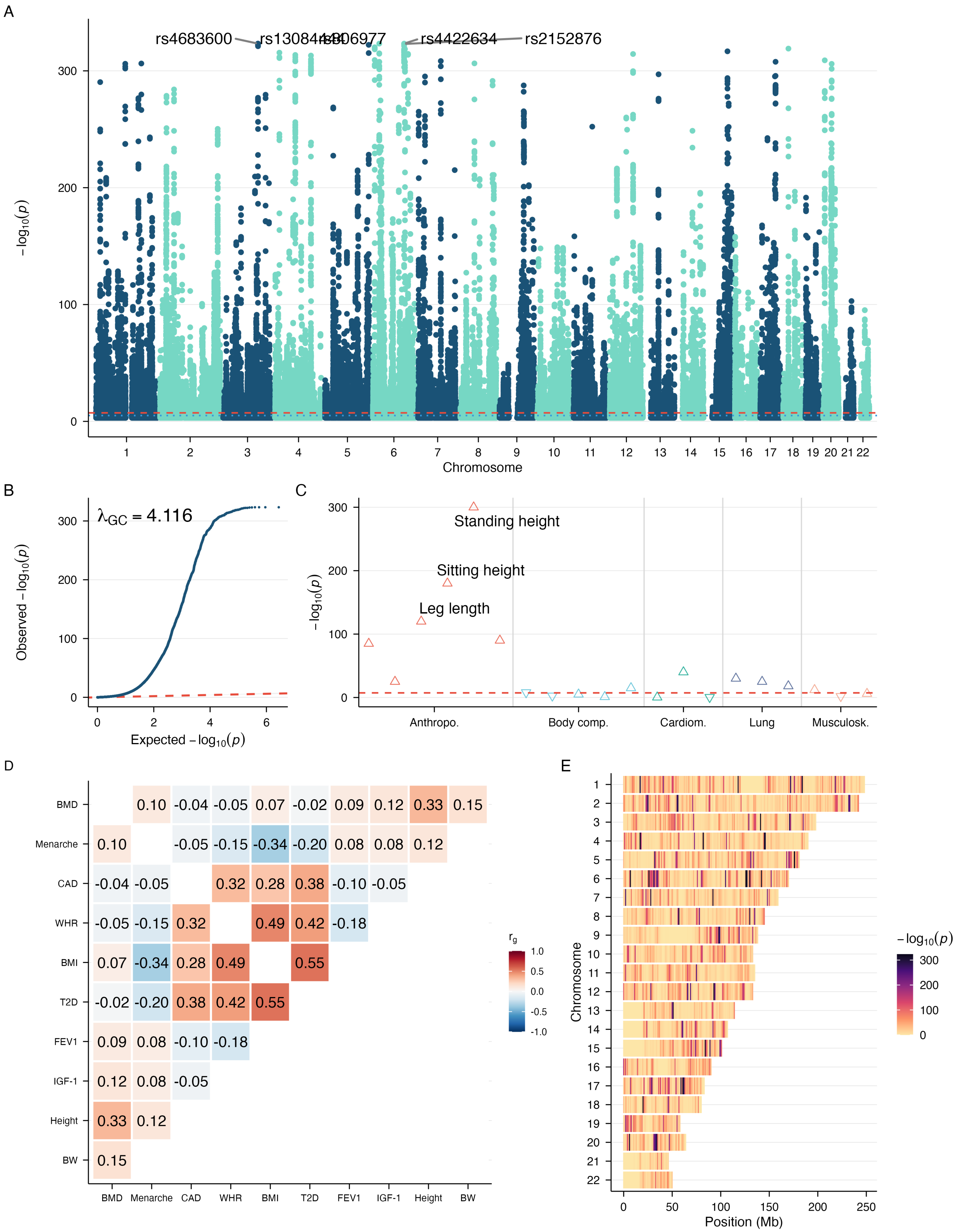

Fig. 1. Overview of ggwas visualizations using GIANT consortium height GWAS summary statistics (Yengo *et al.*, 2022) (1.37M variants, European ancestry). (A) Manhattan plot with labeled top hits. (B) QQ plot with 95% confidence band and λ_{GC}. (C) PheWAS plot showing height-associated phenotypes with effect direction triangles. (D) Genetic correlation matrix (published LD Score regression estimates). (E) Genome-wide p-value heatmap (darker shading indicates stronger association signal).